

Basic Principles of Medicine 1
Module : Foundations to Medicine
Lecture No: 42

**Title: Purine Degradation and
Pyrimidine Metabolism**

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Recommended Reading

- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- **Chapter 22 (Lippincott's Biochemistry 9th edition);** Nucleotide Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library
- **Focused Case 7: Joint pain page 595**
 - Clinical Cases | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library
- Practice questions on Sakai

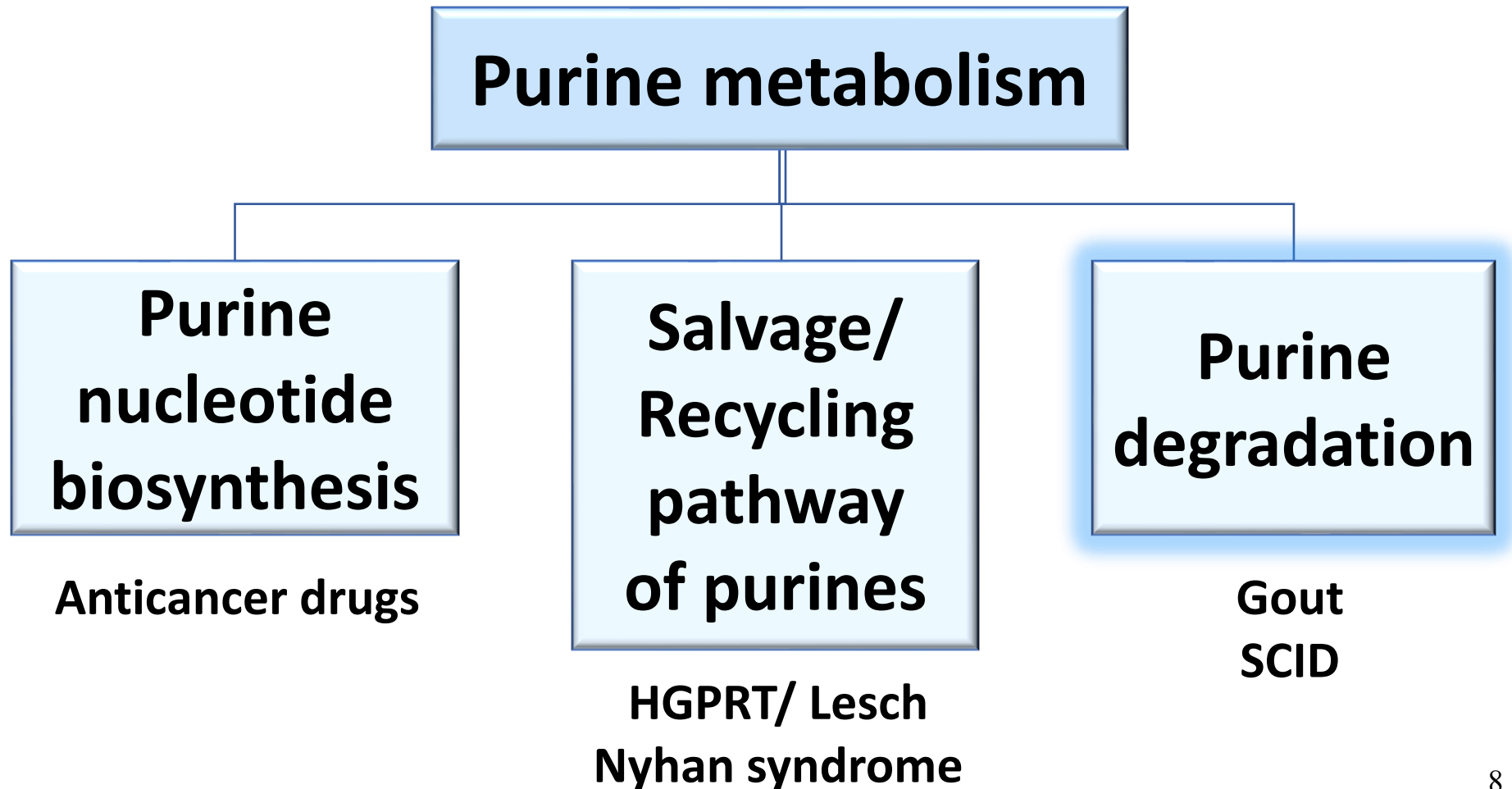
Objectives: Purine Degradation and Pyrimidine Synthesis

SOM.MK.I.BPM1.1.FTM.3.BCHM.0169	Discuss purine nucleotide catabolism and explain uric acid formation
SOM.MK.I.BPM1.1.FTM.3.BCHM.0170	Analyze causes of hyperuricemia (overproduction and underexcretion of uric acid)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0171	Correlate the basis of clinical features and explain basis of use of allopurinol and febuxostat in the management of hyperuricemia
SOM.MK.I.BPM1.1.FTM.3.BCHM.0172	Propose molecular basis of immunodeficiency in autosomal recessive SCID (ADA deficiency) and deficiency of purine nucleoside phosphorylase.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0173	Enumerate the donors of C and N atoms to pyrimidine ring
SOM.MK.I.BPM1.1.FTM.3.BCHM.0174	Outline pyrimidine biosynthesis (UTP, CTP and dTMP). Distinguish purine and pyrimidine biosynthetic pathways
SOM.MK.I.BPM1.1.FTM.3.BCHM.0175	Summarize the regulation of pyrimidine biosynthesis (regulatory enzymes, regulators)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0176	Explain formation of deoxyribonucleotides from ribonucleotides (Specify coenzyme and inhibitor)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0177	Indicate mechanism of action of 5-fluoro uracil and methotrexate and their clinical application
SOM.MK.I.BPM1.1.FTM.3.BCHM.0178	Specify enzyme deficiency and explain metabolic basis of clinical features in orotic aciduria
SOM.MK.I.BPM1.1.FTM.3.BCHM.0195	Compare and contrast the roles and subcellular location of CPS-I (urea cycle) and CPS-II (pyrimidine biosynthesis)

Overview of functions of nucleotides

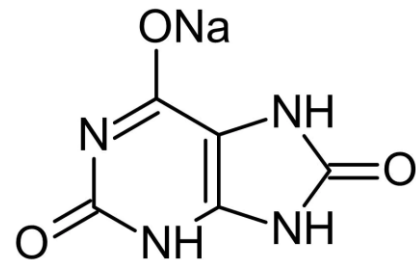
- Ribonucleotides are building blocks of RNA
- Deoxyribonucleotides are building blocks of DNA
- ATP is the energy currency of cells
 - Components of coenzymes like FAD, NAD, NADP and coenzyme A
- Nucleotide triphosphates (NTP) can activate reactions eg. UDP-glucose
- Nucleotides like cAMP and cGMP
 - intracellular second messengers
- ★ • Several anti-cancer drugs and antibacterials are inhibitors of purine and pyrimidine nucleotide metabolism
- ★ • Nucleotide metabolism defects associated with human disorders

Purine nucleotide metabolism



Hyperuricemia and Gout

- A 50-year-old **man** comes to the emergency room with an acutely swollen right **big toe**.
- Sudden onset of **pain and warmth** in the joint. He could not bear weight (severe pain).
- **Diagnosis:** Joint fluid aspiration and microscopic examination revealed presence of **needle shaped monosodium urate crystals** in synovial fluid



What is uric acid??

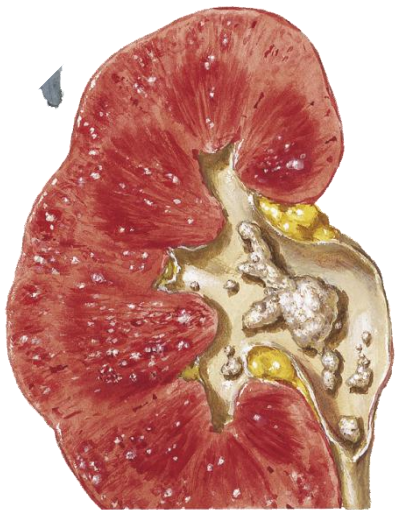


Tophi in hand

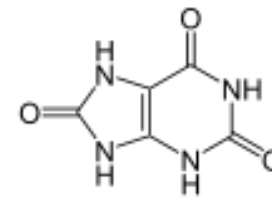


Tophi in auricle

- End product of **purine breakdown**
 - Adenine, guanine, hypoxanthine
- Normally excreted in urine
- **Low solubility**
 - High blood uric acid levels (hyperuricemia),
 - **Monosodium urate** precipitates in joint (synovial) fluid causing arthritis (**gouty arthritis**)
- **Monosodium urate** deposits in soft tissues as '**tophi**'
- **Renal stones** of uric acid

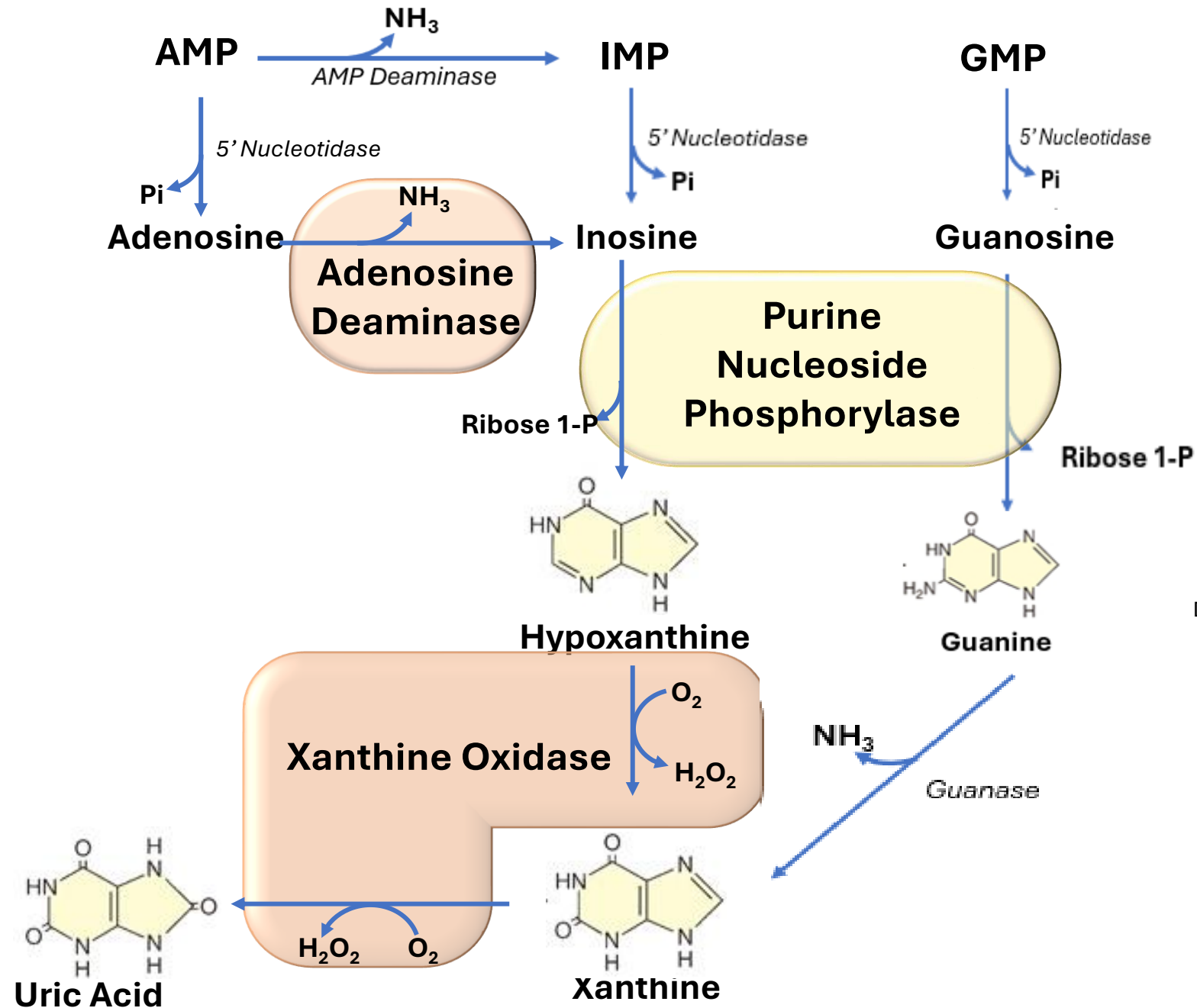


Urate stones in renal pelvis



Before and after 27 months treatment

Degradation of Purine Nucleotides to Uric Acid



Do NOT memorize Chemical formula

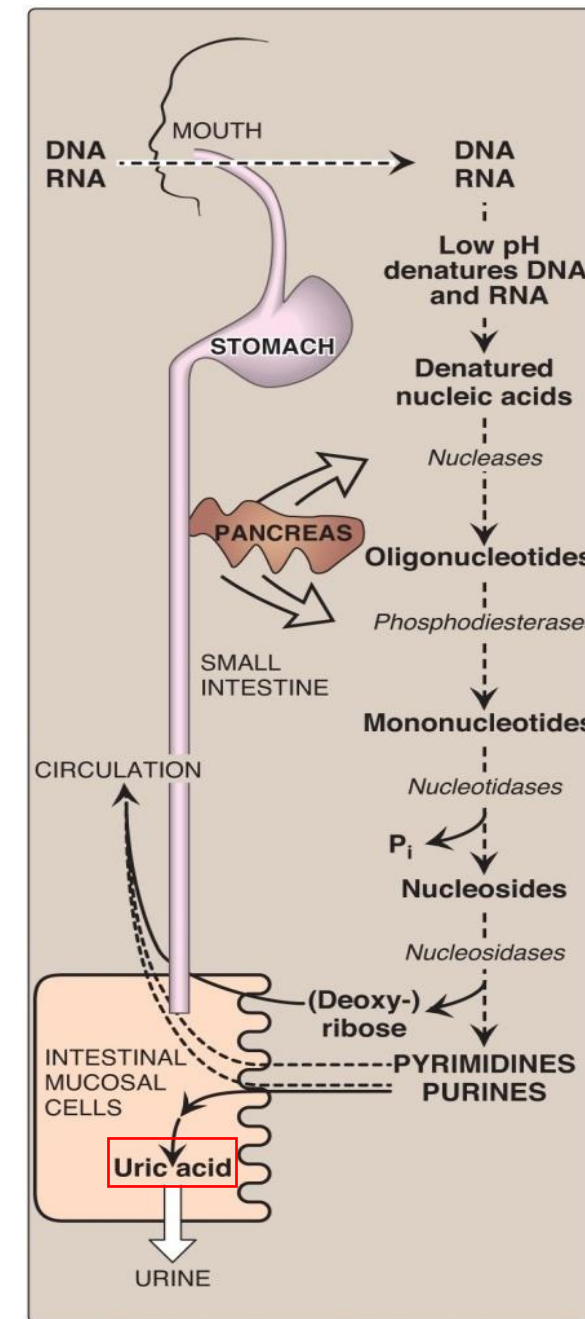
Formation of Uric acid

- AMP (nucleotide) converted to adenosine (nucleoside)
- **Adenosine** (adenine-ribose) is deaminated (removal of ammonia) to form **inosine** (hypoxanthine-ribose) by **adenosine deaminase (ADA)**
- Ribose removed by **purine nucleoside phosphorylase (PNP)** to form hypoxanthine (purine base)
- Hypoxanthine → **Xanthine** → **Uric acid** by **xanthine oxidase**
- Guanine → xanthine by guanase and later forms uric acid by xanthine oxidase
- End product of **purine** degradation (adenine, hypoxanthine and guanine)
- Blood Uric acid levels are close to saturation limits!!



Dietary nucleic acids and purine bases

- Dietary nucleic acids (DNA and RNA) in **meat/ beer** digested to free purine bases in intestine
- Dietary purine bases → **Uric acid** and excreted in urine
- Dietary purine bases not reused by salvage pathway
- Hyperuricemia: Modify diet and reduce nucleic acid (dietary purine content) – Avoid meat and beer
- Note: End product of amino acid metabolism is Urea



Hyperuricemia: **Underexcretion**

- Uric acid levels in blood close to saturation point
- Blood uric acid levels >6.8 mg/dL
- **Monosodium urate** crystals in joints and soft tissues (Gout)
- Hyperuricemia may be caused by
 - **Underexcretion** of uric acid (more common)
 - Cause not defined – males and obesity (risk factors)
 - Renal disease
 - Lactic acidosis (urate competes with lactate for excretion)
 - Inherited disorders: Von Gierke disease, hereditary fructose intolerance

Hyperuricemia: Overproduction of uric acid

- Idiopathic (cause not known): Males and obesity (risk factors)



- **PRPP synthetase gene** mutations

- $\uparrow V_{\max}$, $\downarrow K_m$ (Gain-of function) = **Overactive enzyme**
- Elevated [**PRPP**] (Activates purine synthesis)
- Excessive turnover of purines



- **HGPRT Deficiency** (**Lesch-Nyhan syndrome**)

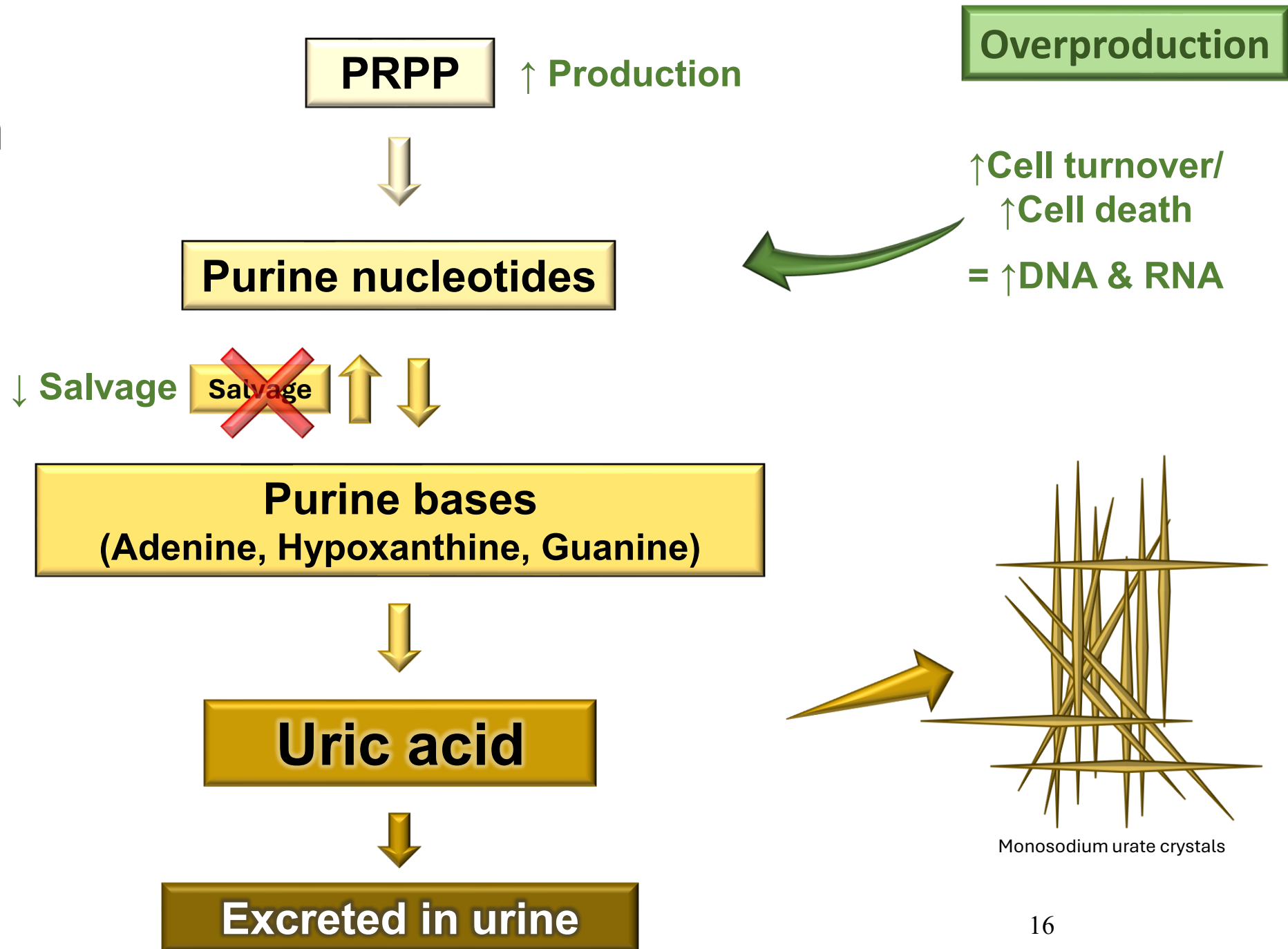
- **Less recycling (salvage)** of purine bases and **elevated [PRPP]**
- Stimulates purine biosynthesis
- Increased Purine turnover

- Patients on **chemotherapy** (Increased cell death)

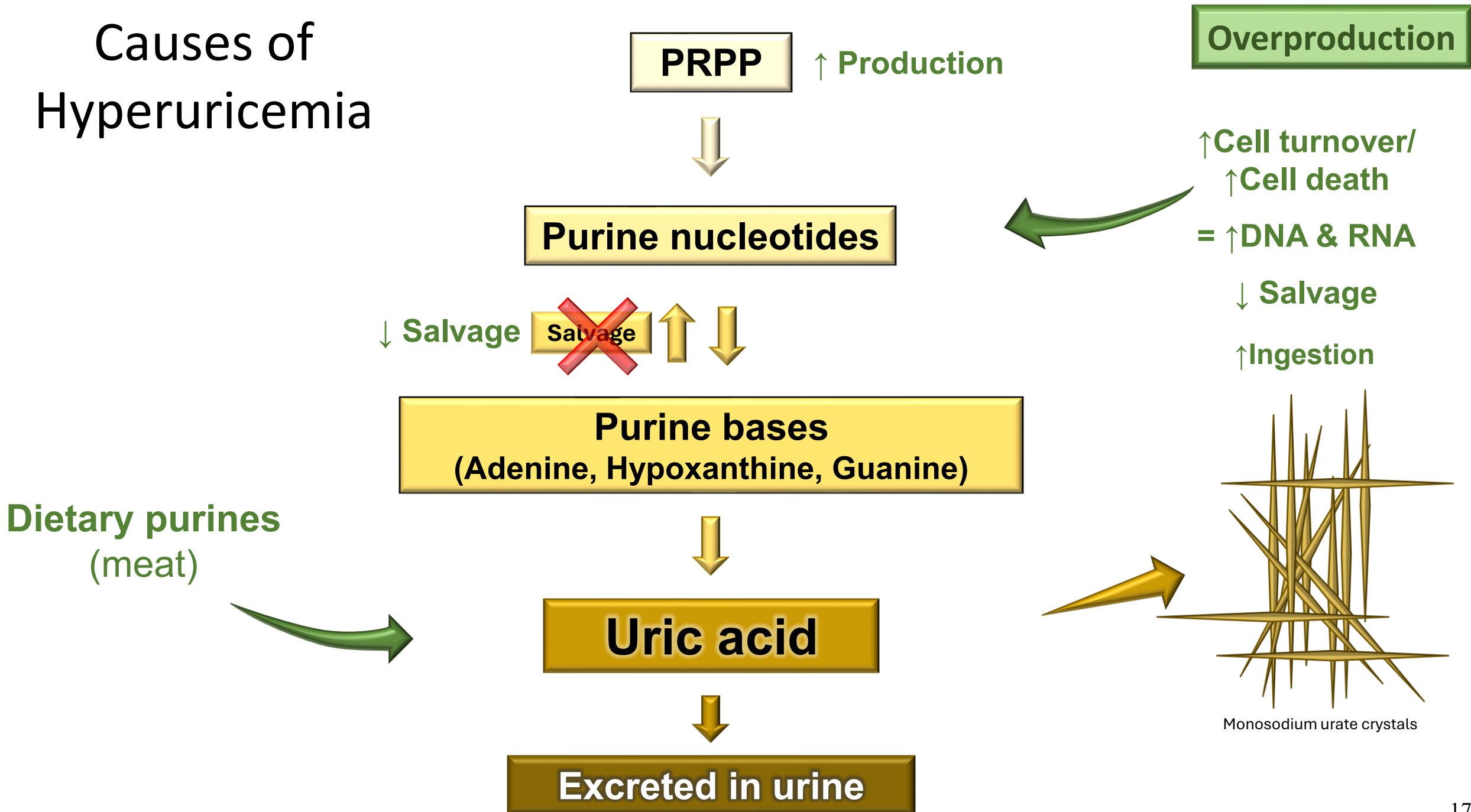
- Increased DNA and RNA breakdown

- Inherited disorders: Von Gierke disease, hereditary fructose intolerance

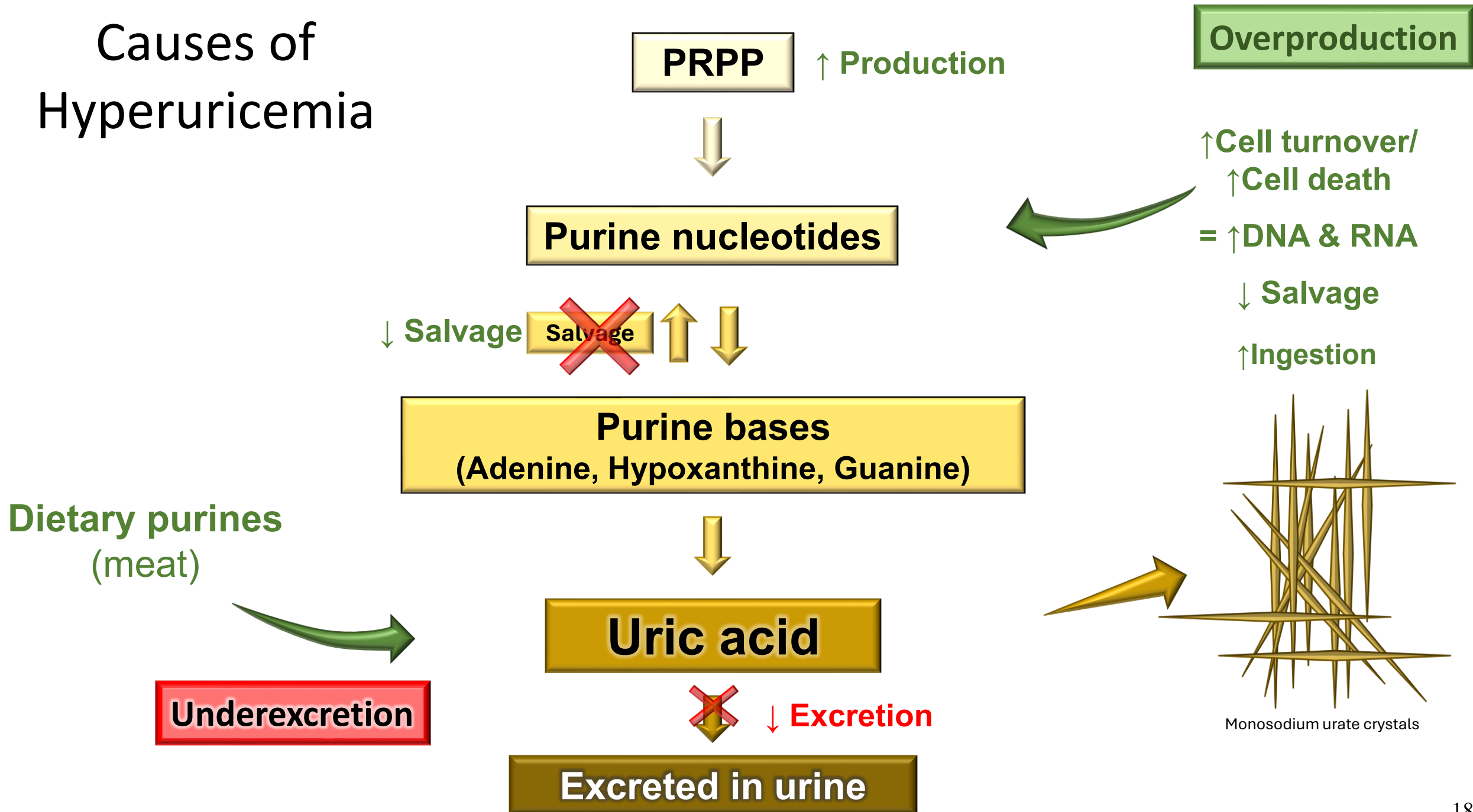
Causes of Hyperuricemia



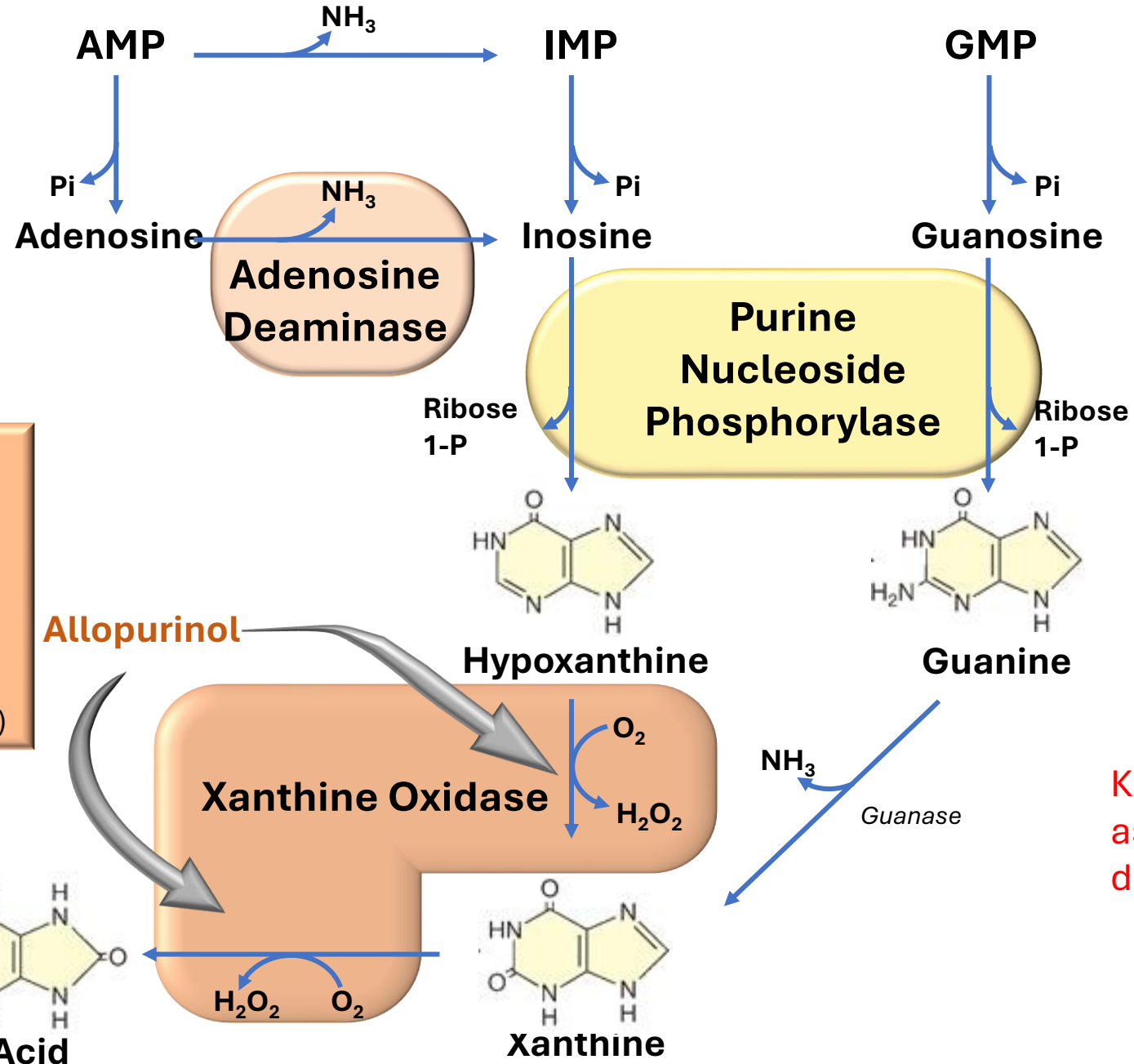
Causes of Hyperuricemia



Causes of Hyperuricemia



Hyperuricemia: Gout



Know the reactions associated with disorders or inhibitors



Gout

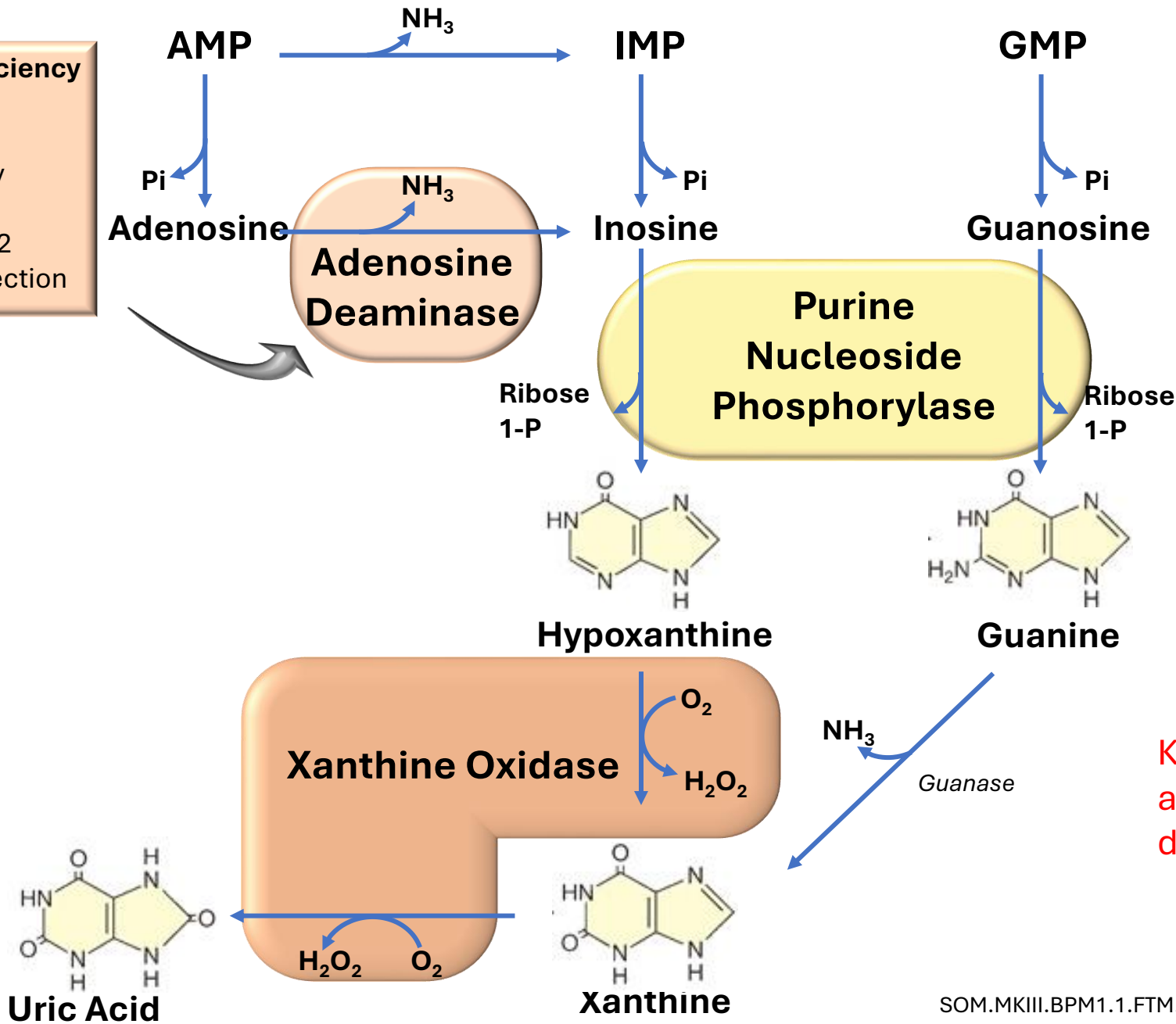
Hyperuricemia: Uric acid $\rightarrow$ Monosodium urate crystals $\rightarrow$ Recurrent arthritis
Crystals in soft tissues & renal stones
Treatment: **Allopurinol** inhibits xanthine oxidase $\rightarrow$ $\uparrow$ hypoxanthine & xanthine (more soluble than uric acid)

Management of hyperuricemia

- Colchicine and anti-inflammatory agents (NSAID and cortisol) reduce pain and inflammation
- Urate lowering therapy (lower uric acid levels)
 - **Allopurinol** : Inhibits **Xanthine oxidase** and decreases uric acid formation
 - Allopurinol (hypoxanthine analog) converted to oxypurinol
 - Suicide inhibitor
 - Hypoxanthine and xanthine more soluble than urate
 - **Febuxostat** (Xanthine oxidase inhibitor)
- Increase uric acid excretion (**uricosuric agents**) (probenecid)
- **Decrease meat** and **alcohol**
 - Dietary purine bases form uric acid

ADA Deficiency: SCID

Adenosine Deaminase Deficiency (ADA)
Autosomal recessive severe combined immunodeficiency (SCID)
Untreated usually die before 2 years from overwhelming infection



Know reactions associated with disorders



Adenosine deaminase deficiency (AR-SCID)

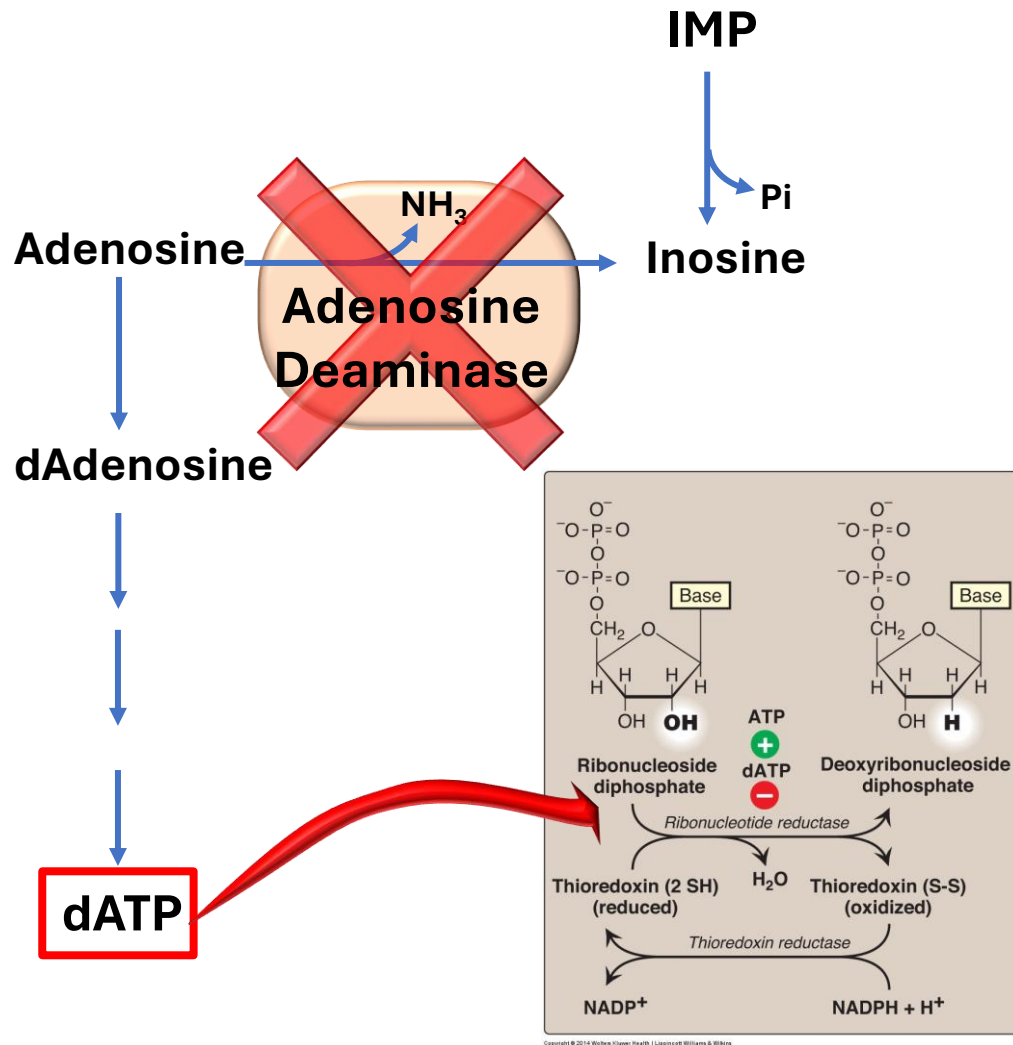
- A 4-month-old male/female with a 2-month history of persistent cough
- Antibiotics but no improvement
- Respiratory distress due to **pneumonia**
- Serum **immunoglobulin** levels **very low**
- **T and B cell function** severely compromised
- Autosomal recessive SCID
 - (there is an X-linked form as well)



<https://www.cbsnews.com/pictures/bubble-boy-40-years-later-look-back-at-heartbreaking-case/>

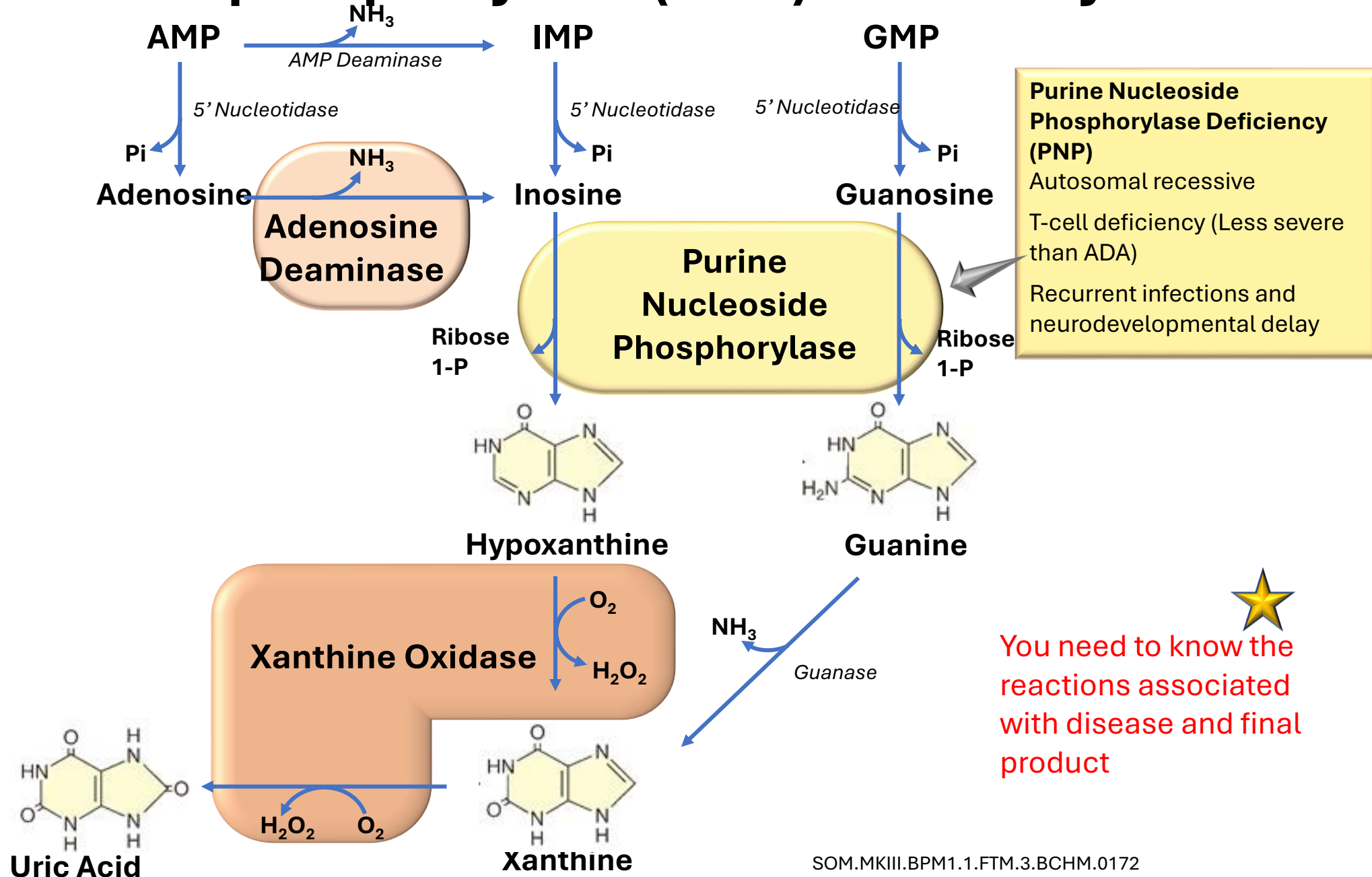
<https://www.youtube.com/watch?v=B84GJOsioSA>

AR-SCID: Adenosine Deaminase (ADA) deficiency



- Adenosine accumulates
 - → Increased [dATP] levels
 - → Inhibits ribonucleotide reductase
 - → Reduced lymphocyte cell division
- Decrease **T-cell and B-cell immunity**
- Recurrent infections; Fatal, life-threatening
- Treatment: **Enzyme replacement** or bone marrow transplant
- Gene therapy successful
- [Twenty-Five Years of Gene Therapy for ADA-SCID: From Bubble Babies to an Approved Drug](#)

Purine nucleoside phosphorylase (PNP) deficiency



Purine nucleoside phosphorylase (PNP) deficiency

- Inherited deficiency of [PNP](#) (purine degradation) less common
- Less severe immunodeficiency (than ADA) involving T-cells
- Repeated infections; developmental delay
- Lab tests indicate **T-cell deficiency**
- **Therapy:**
 - Hematopoietic stem cell transplant

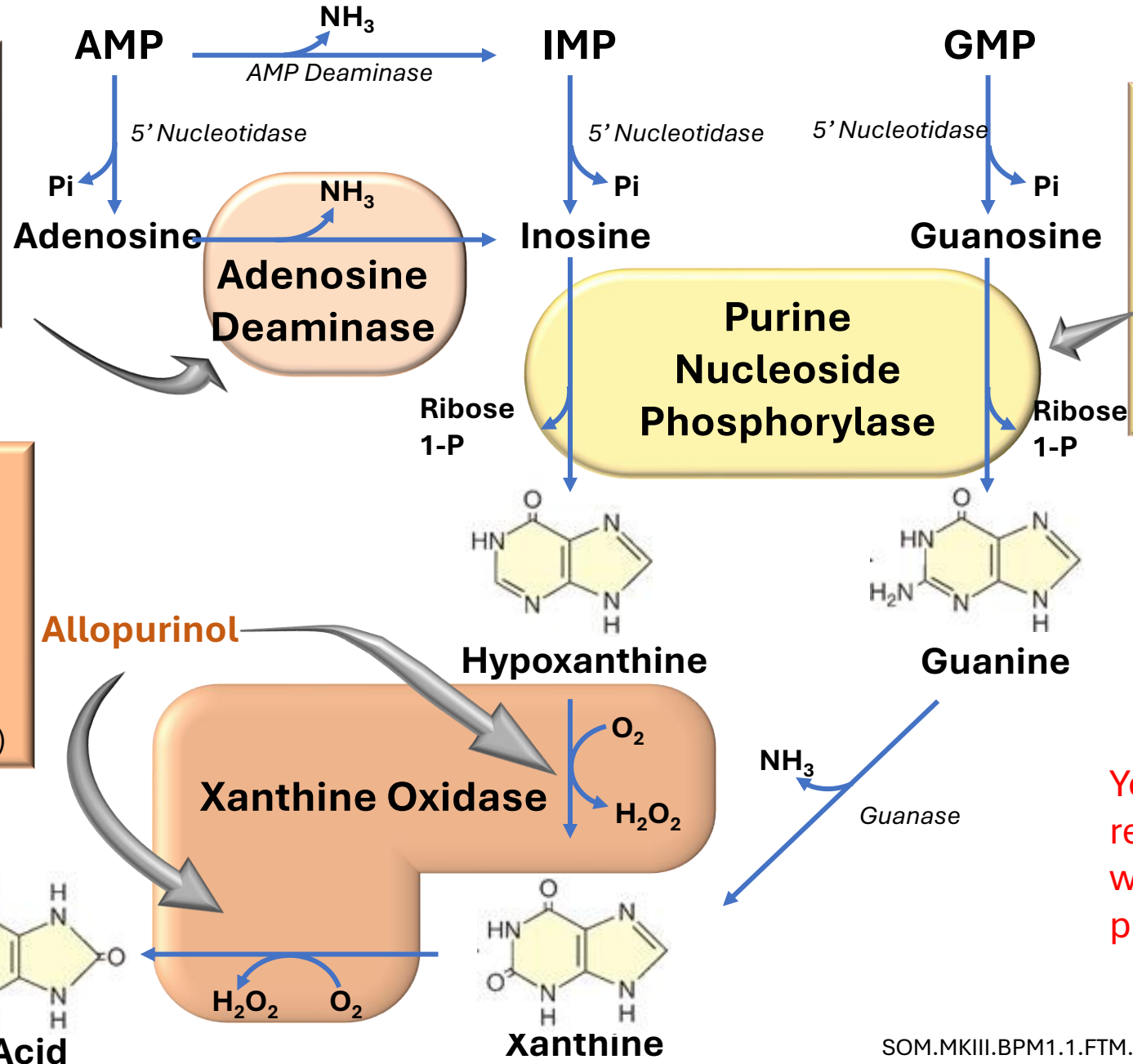
Summary

Adenosine Deaminase Deficiency (ADA)

Autosomal recessive severe combined immunodeficiency (SCID)
Untreated usually die before 2 years from overwhelming infection

Gout

Hyperuricemia: Uric acid → Monosodium urate crystals → Recurrent arthritis
Crystals in soft tissues & renal stones
Treatment: **Allopurinol** inhibits xanthine oxidase → ↑ hypoxanthine & xanthine (more soluble than uric acid)



Purine Nucleoside Phosphorylase Deficiency (PNP)

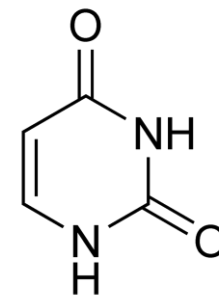
Autosomal recessive
T-cell deficiency (Less severe than ADA)
Recurrent infections and neurodevelopmental delay

You need to know the reactions associated with disease and final product

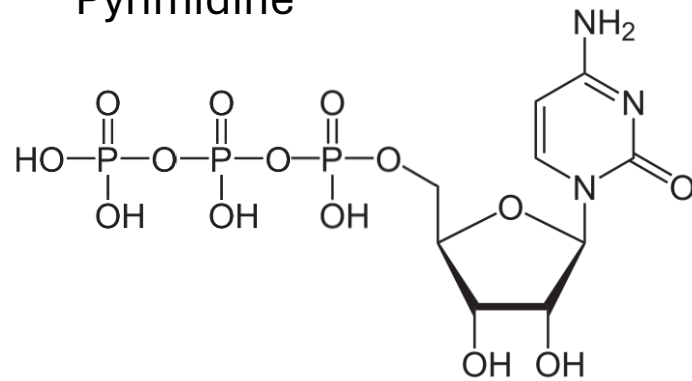




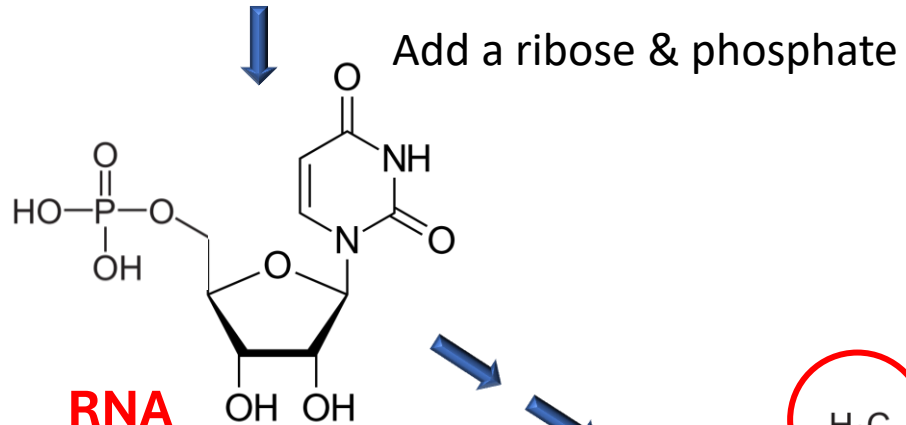
Pyrimidine



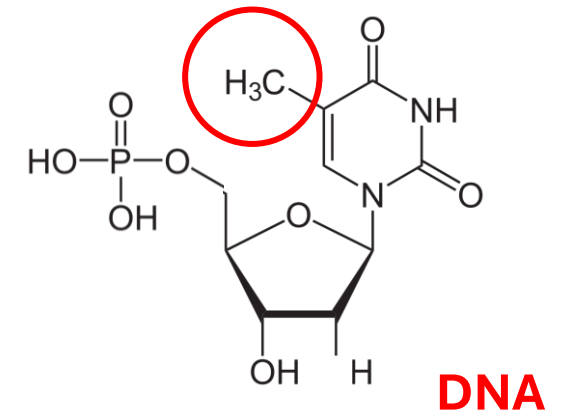
Uracil



Cytidine triphosphate (CTP)



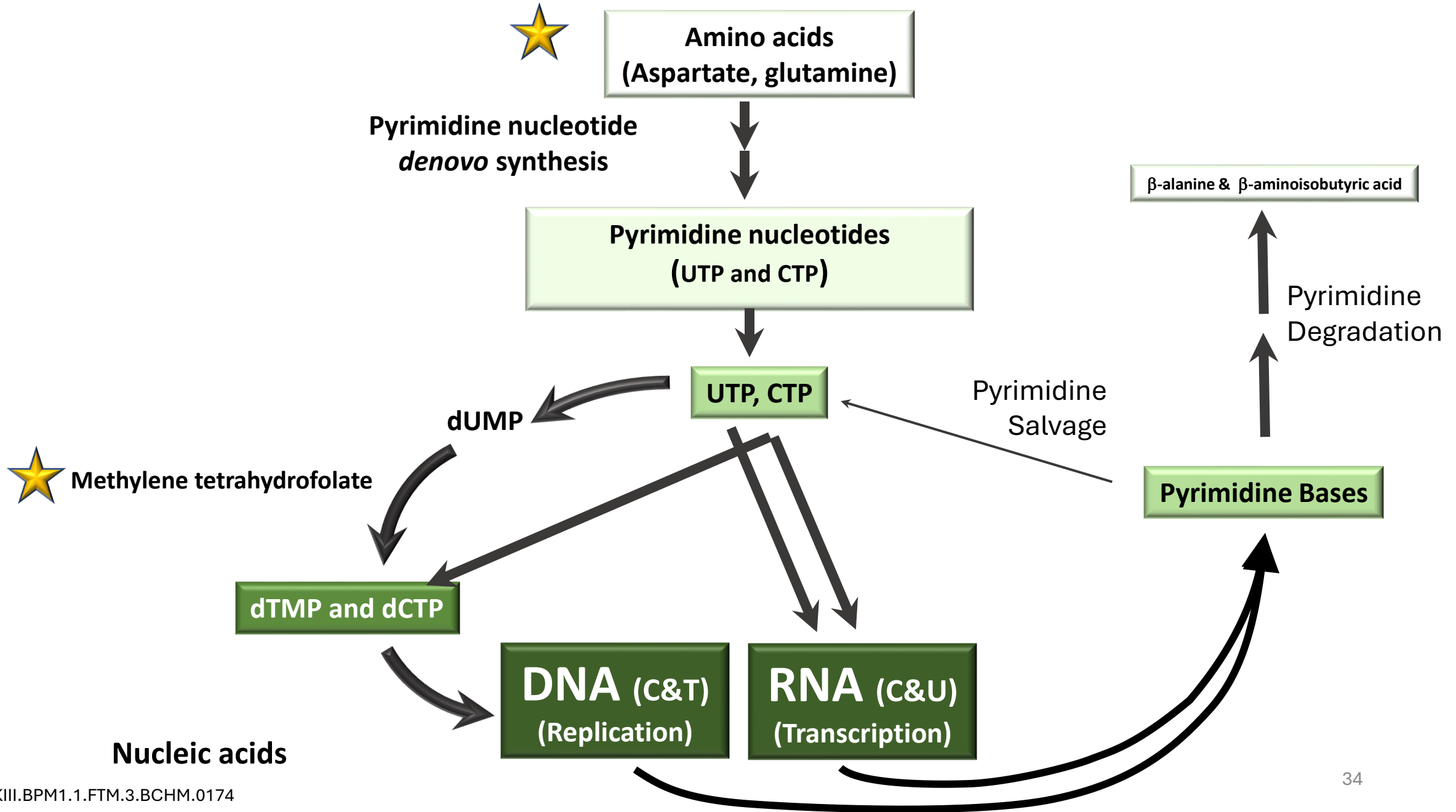
RNA
Uridine monophosphate (UMP)



DNA
Deoxythymidine monophosphate (dTMP)

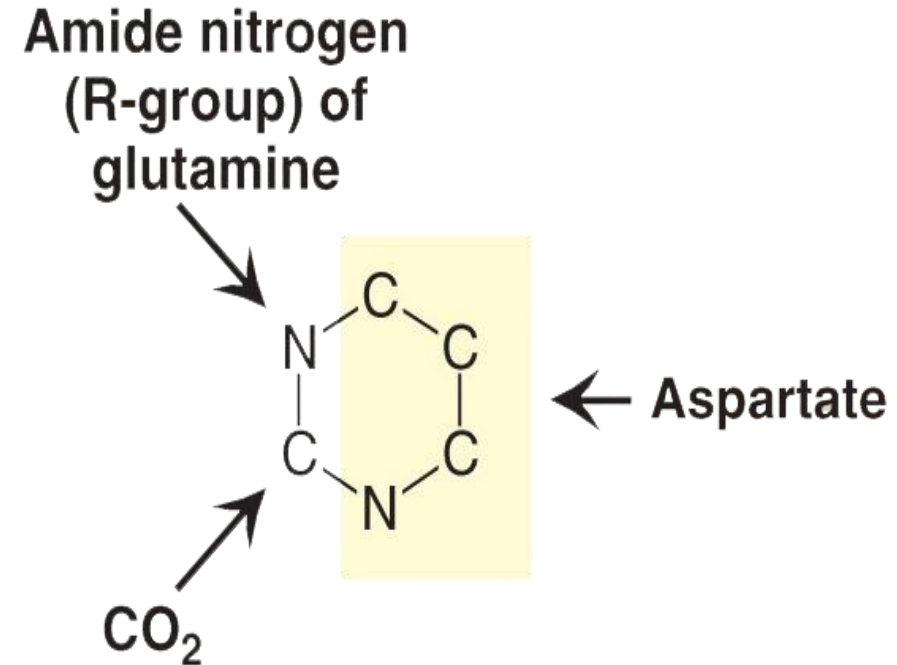
Pyrimidine Nucleotide Metabolism

Overview of pyrimidine nucleotide metabolism

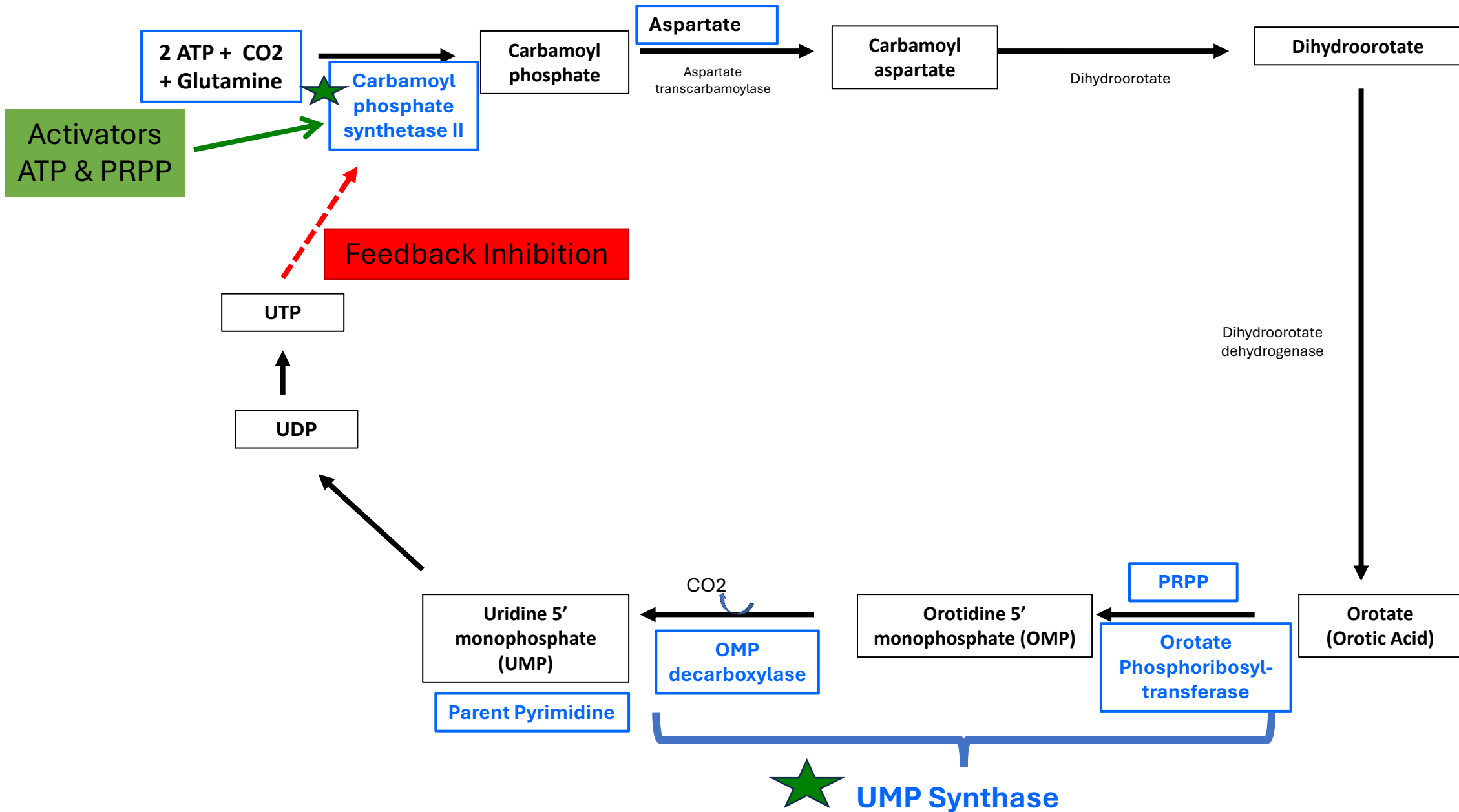


Closer Look: Pyrimidine Synthesis

- Synthesis of pyrimidines bases Uracil (RNA), Cytosine and Thymine (DNA)
- Precursors: Aspartate, CO_2 , and glutamine
- Pyrimidine ring initially synthesized **before attachment** to ribose-5-phosphate, which is donated by PRPP (compare to purine biosynthesis)
- No folic acid is required to make uridine and cytidine
- **HOWEVER**, synthesis of thymidine requires 1-carbon group from **methylene tetrahydrofolate (THF) (vitamin B9/folic acid)** ★



De novo Pyrimidine Synthesis

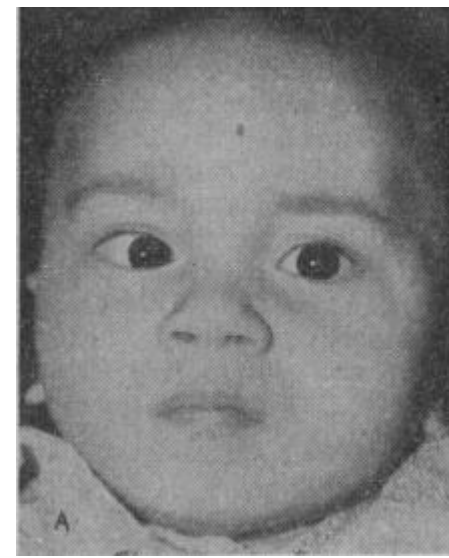


Pyrimidine biosynthesis and regulation

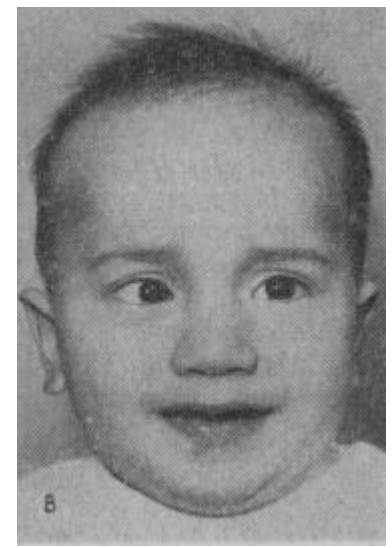
- **CPS-II** (carbamoyl phosphate synthetase-II): Regulatory step of pyrimidine biosynthesis (compare to CPS I of urea cycle)
- CPS-II **inhibited by UTP** (end-product) by **feedback** inhibition
- CPS-II **activated by ATP** (ensures equal concentration of purine and pyrimidine nucleotides) and **PRPP**

Case Report: Orotic Aciduria

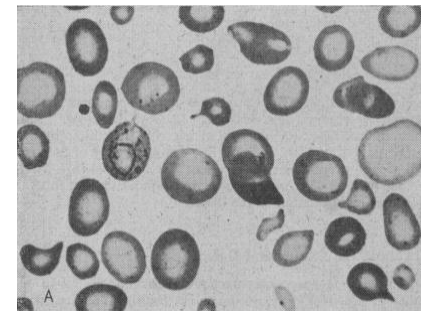
- A 17-month-old infant is weak and anemic
- Macrocytic, megaloblastic anemia, resistant to iron, B12, B6, folic acid (B9)
- Height and weight of child less than normal
- Orotic acid excretion in urine
- Normal blood ammonia levels



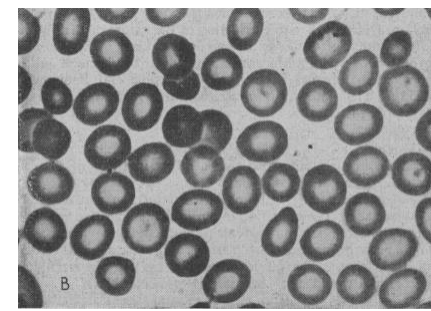
At diagnosis:
Developmental delay,
weight at 3rd and height at
10th centile, not interactive,
megaloblastic anemia



After treatment:
Interacts with
surroundings, weight
90th and height 50th
centile; No anemia

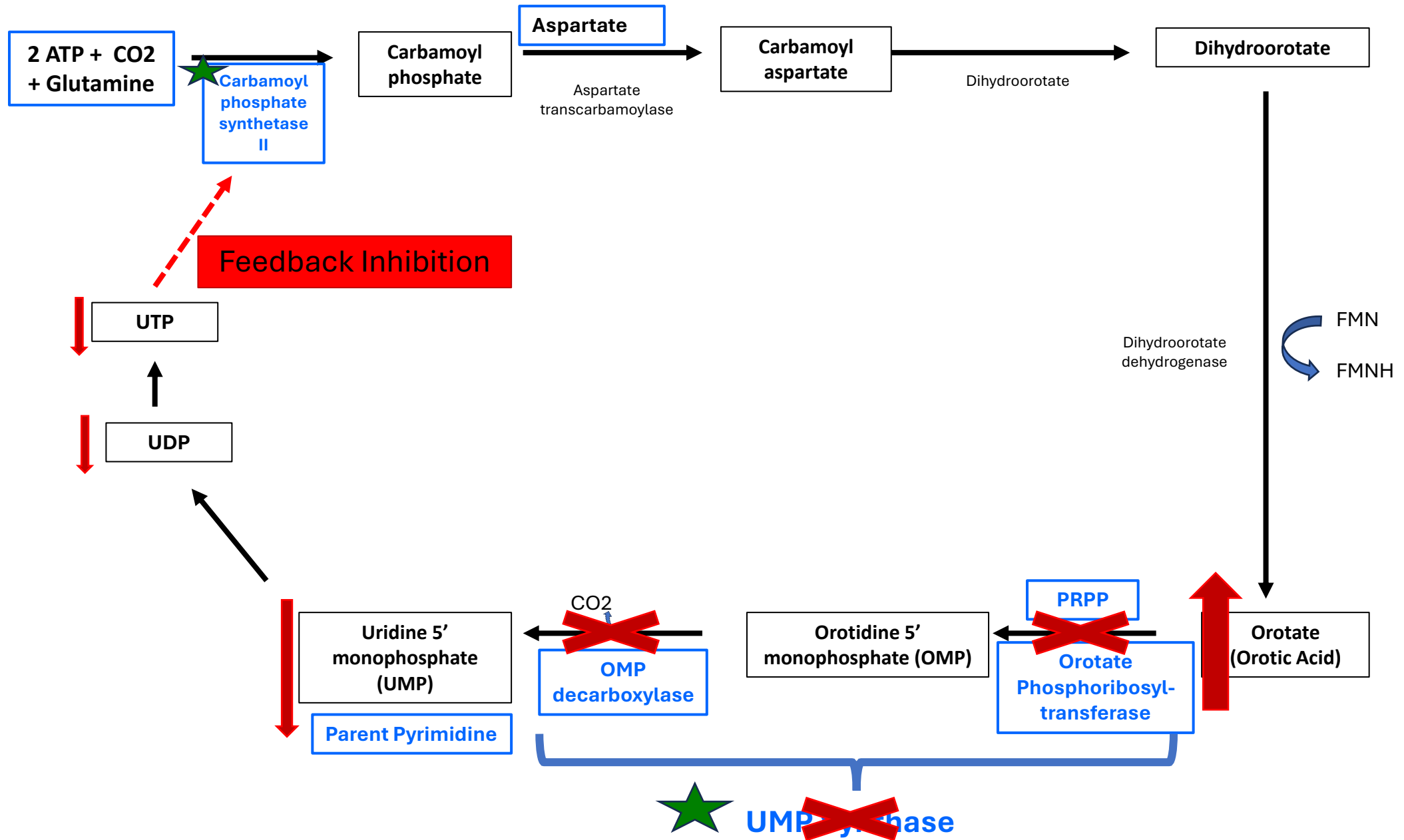


Before treatment



After **uridine**
treatment

Orotic aciduria: Pyrimidine biosynthesis defect



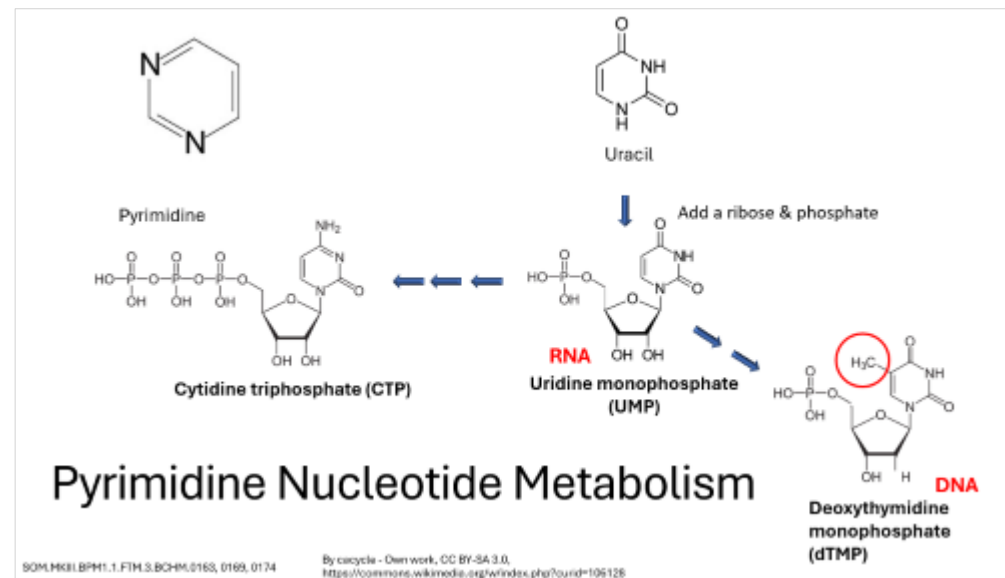
Orotic aciduria: Pyrimidine biosynthesis defect

- **OPRT and OMP decarboxylase:** Both activities present on same protein. Also known as **UMP synthase**
- Deficiency of **OPRT and/or OMP decarboxylase (UMP synthase)**
- Elevated **orotic acid levels** in blood and urine
- Pyrimidine deficiency results in **decreased RBC formation**; megaloblastic anemia, reduced rates of DNA synthesis and cell division
- **Treatment: Uridine Therapy**
- **Uridine** improves anemia
 - Common precursor for synthesis of **thymidine and cytidine**
- UTP **feedback inhibitor of CPS-II** and lowers orotic acid levels

Differentiate from Orotic aciduria due to Urea cycle defect

- Orotic aciduria is also found in **Hyperammonemia Type II** (deficiency of ornithine transcarbamoylase of urea cycle)
- Increased amounts of **carbamoyl phosphate** formed in the mitochondria (urea cycle), diffuses into **cytosol** and is used for pyrimidine biosynthesis, causing increased **formation of orotic acid and orotic aciduria**

Biochemical Pathway	Pyrimidine Biosynthesis	Urea cycle
Diagnosis	Orotic Aciduria	Hyperammonemia Type II
Enzyme Deficiency	UMP Synthase/OPRT/OMP decarboxylase	OTC
Regulated enzyme	CPS II	CPS I
Cellular location	Cytosol	Mitochondria
Metabolite elevated	Orotic acid	Orotic acid + Ammonia
Clinical features	Macrocytic anemia	Neurological findings: Seizures



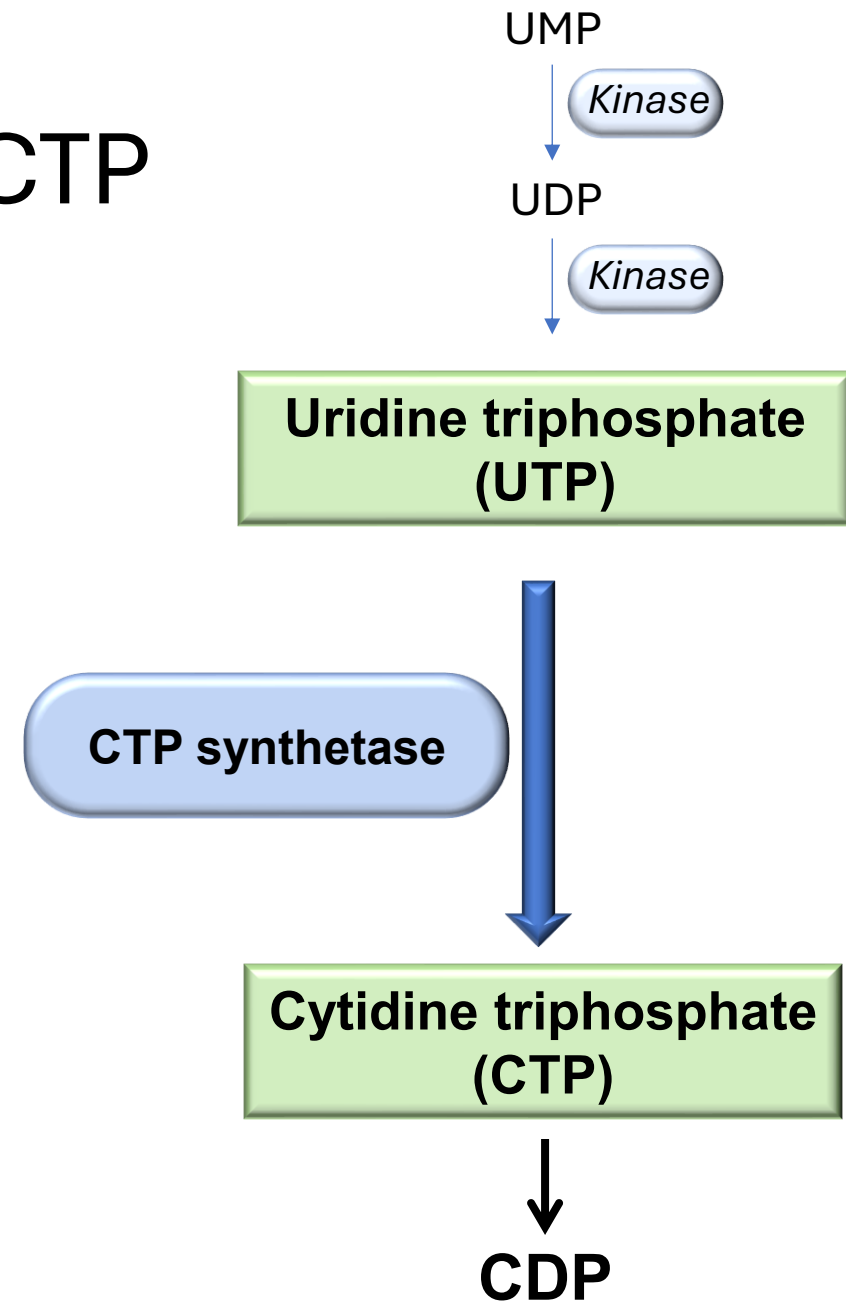
UMP is the Parent pyrimidine

From UMP we make CDP

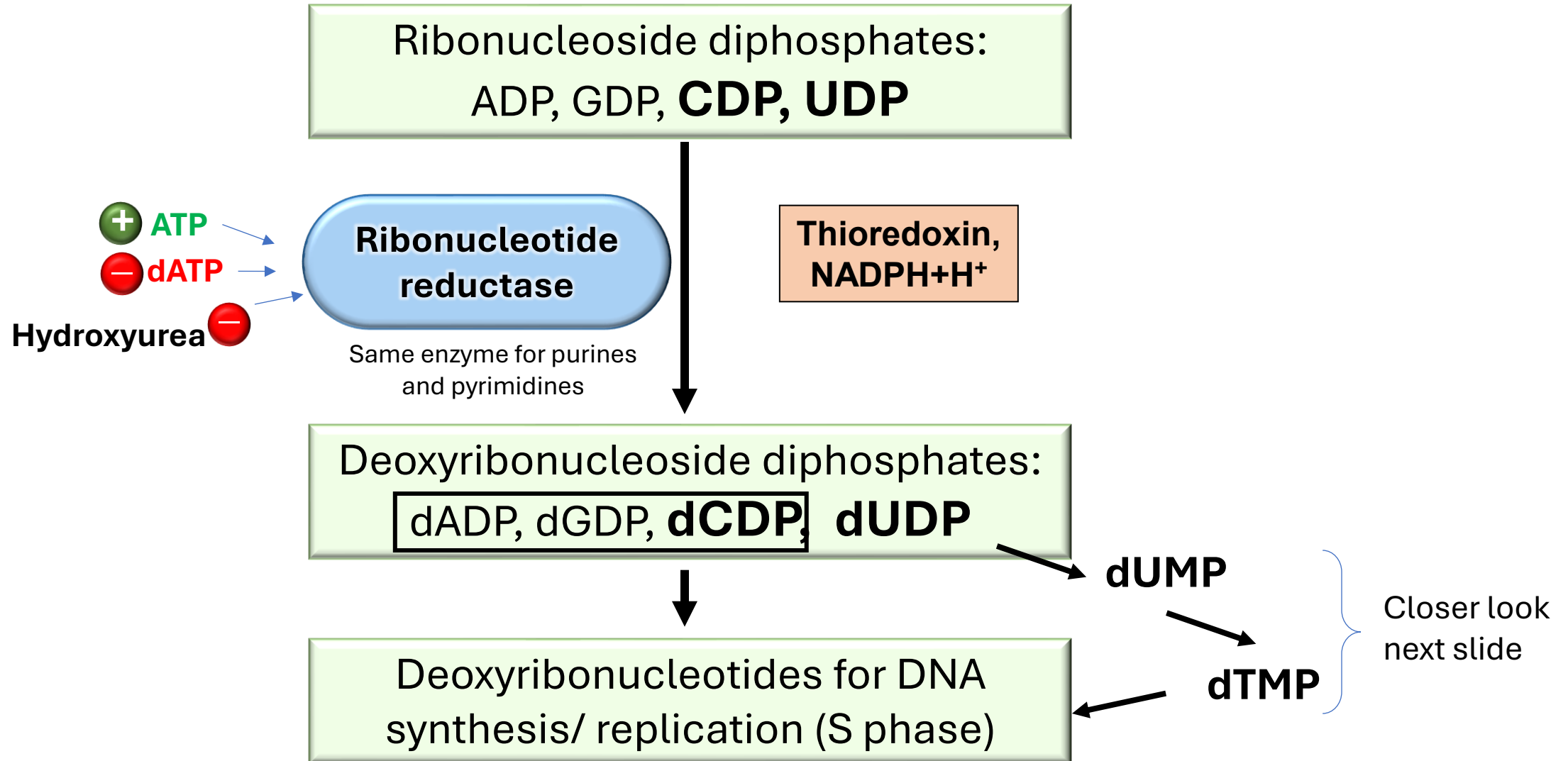
From dUMP we make dTMP

Closer Look: Formation of CTP

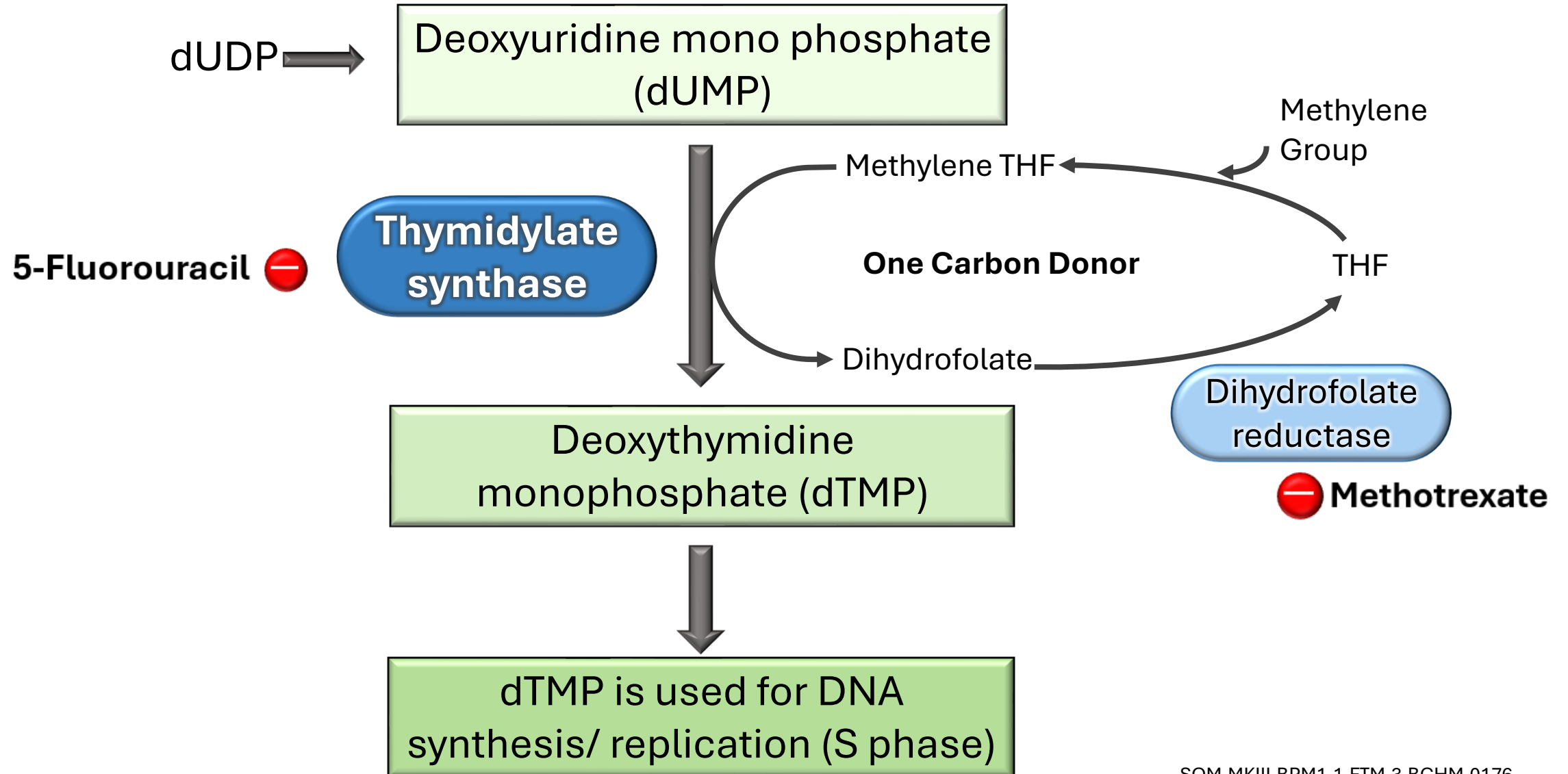
- UTP forms CTP by **CTP synthetase**
- Uracil and cytosine (UDP and CDP) synthesized in **ribo**nucleotide form
- CDP and UDP are converted to **deoxy**ribonucleotides by **ribonucleotide reductase**



Closer Look: Formation of dNTPs (deoxyribonucleotides)



Closer look: Synthesis of dTMP & Inhibitors



Thymidine synthesis

- Thymidine monophosphate synthesized from dUMP by **thymidylate synthase**
- Remember: DNA does not have Uracil
- Requires **methylene tetrahydrofolate (THF)/ folic acid/ vitamin B9 – one carbon donor**
- Methylene tetrahydrofolate converted to **dihydrofolate**
- Dihydrofolate converted THF by **dihydrofolate reductase**
- dTMP: ONLY pyrimidine that needs methylene tetrahydrofolate (vitamin B9)

Anticancer drugs (Inhibitors of thymidine synthesis)

- **5 Fluorouracil** converted to 5 fluoro-dUMP
 - Irreversibly binds **thymidylate synthase** and inactivates it (suicide inhibitor) (refer: Enzymes)
- **Methotrexate** (folate analog) competitively inhibits **dihydrofolate reductase** and prevents THF formation (Methotrexate inhibits **purine and thymidine** biosynthesis)



Folic acid (folate/ vitamin B9): Requirement and deficiency

- Formyl-THF or methylene-THF used for **purine nucleotide and thymidine synthesis** (dTMP synthesis)
- Essential for DNA synthesis during S phase of cell cycle
- Critical for rapidly dividing cells
- Folate deficiency: **Macrocytic anemia** and decreased cell division; **Impaired purine and pyrimidine nucleotide synthesis**
- Pre and perinatal folate deficiency in pregnancy
 - Higher risk of fetal **neural tube defects**



Pyrimidine salvage and degradation

- Pyrimidine bases recycled to their nucleotides by enzymes using PRPP
- Pyrimidine bases catabolized β -alanine & β -aminoisobutyrate (water soluble end products), with formation of ammonia and CO₂

Fill in the table

Pharmacologic agents	Enzymes they inhibit
Allopurinol/ Febuxostat	
Methotrexate	
Sulfonamide/ sulfa drugs	
Trimethoprim	
5-fluorouracil	
Hydroxyurea	
Mycophenolic acid	
NSAIDs	

Answers

Pharmacologic agents	Enzymes they inhibit and mechanism of action
Allopurinol/ Febuxostat	Xanthine oxidase inhibitor (Inhibits conversion of hypoxanthine to xanthine and xanthine to uric acid)
Methotrexate	Eukaryotic dihydrofolate reductase inhibitor (Used as an anticancer agent). Inhibits purine and thymidine synthesis (indirect via inhibition of dihydrofolate reductase)
Sulfonamide/ sulfa drugs	Inhibit bacterial folate synthesis (used as an antibacterial agent)
Trimethoprim	Bacterial dihydrofolate reductase inhibitor (Used as an antibacterial agent)
5-fluorouracil	Inhibits thymidylate synthase that converts dUMP to dTMP. This reaction also requires methylene tetrahydrofolate
Hydroxyurea	Inhibits ribonucleotide reductase (conversion of ribonucleotides to deoxyribonucleotides)
Mycophenolic acid	Inhibits the conversion of IMP to GMP (purine nucleotide synthesis); Inhibits IMP dehydrogenase
NSAID	Inhibit cyclooxygenase and used as anti-inflammatory agents

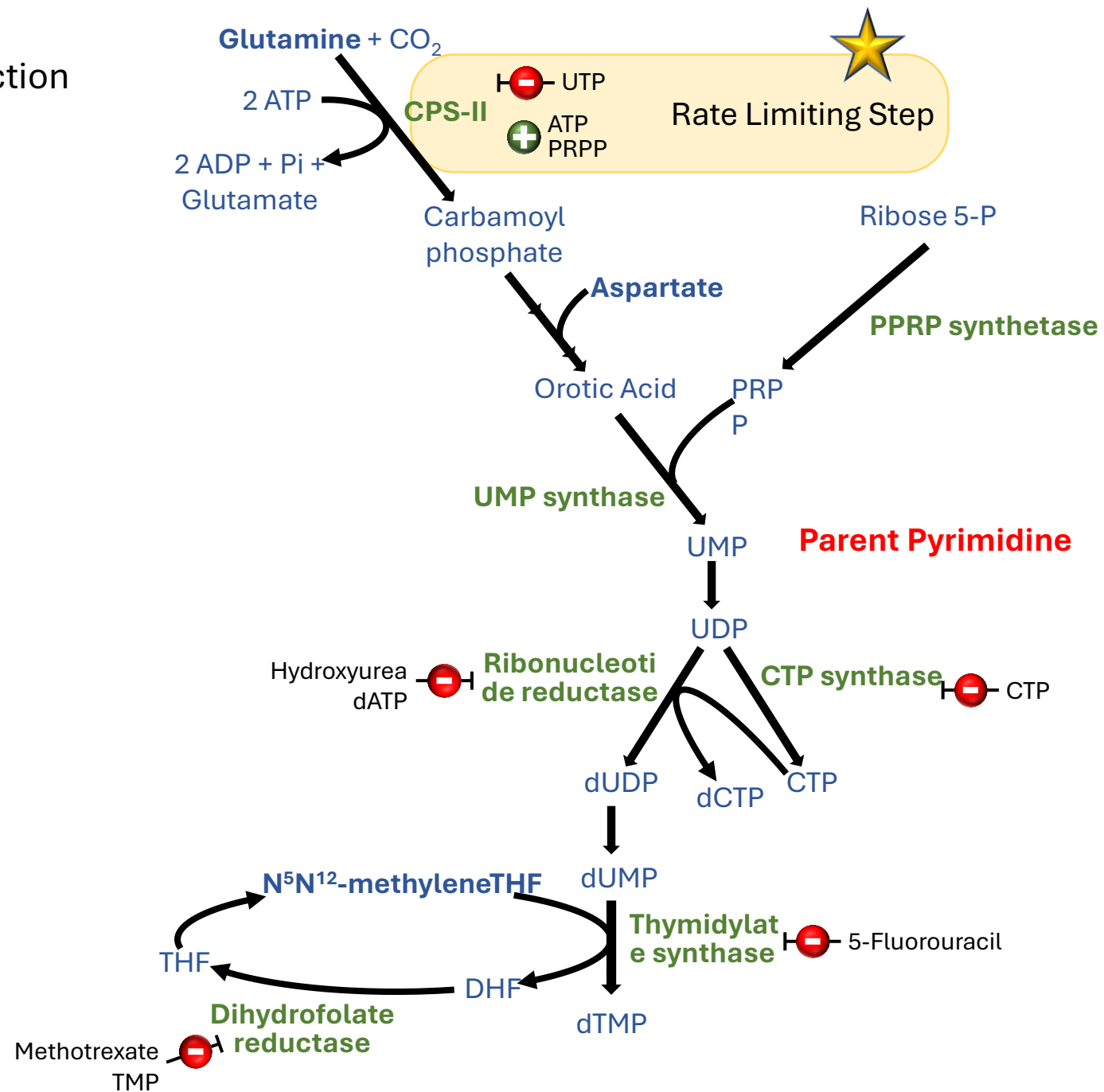
Fill in the table

Enzymes	Functions
PRPP synthetase	
Phosphoribosyl amidotransferase	
Xanthine oxidase	
HGPRT	
CPS-II	
UMP synthase (OPRT and OMP decarboxylase)	
Ribonucleotide reductase	
Thymidylate synthase	
Adenosine deaminase (ADA)	
Purine nucleoside phosphorylase (PNP)	

Answers

Enzymes	Functions
PRPP synthetase	Synthesis of PRPP for purine and pyrimidine nucleotide synthesis
Phosphoribosyl amidotransferase	Regulated/ committed enzyme of purine nucleotide synthesis; Activators: PRPP; Inhibitors: ATP and GTP
Xanthine oxidase	Enzyme of purine degradation; Forms uric acid from xanthine; Also forms xanthine from hypoxanthine; Inhibitor: Allopurinol
HGPRT	Enzyme of Purine salvage pathway; Deficiency causes Lesch Nyhan syndrome
CPS-II	Regulated enzyme of pyrimidine biosynthesis (Inhibited by UTP; Activated by PRPP); Cytosolic enzyme
UMP synthase (OPRT and OMP decarboxylase)	Enzyme of pyrimidine biosynthesis; deficiency of the enzyme results in Orotic aciduria
Ribonucleotide reductase	Converts ribonucleotides to deoxyribonucleotides; Inhibitor: Hydroxyurea
Adenosine deaminase (ADA)	Converts adenosine to inosine in purine degradation pathway. Deficiency results in SCID (T and B cell deficiency)
Purine nucleoside phosphorylase (PNP)	Converts purine nucleoside to free purine base in purine degradation pathway. Deficiency results in T-cell immunodeficiency

Pyrimidine Base Production



De novo Pyrimidine Synthesis

Activators
ATP & PRPP

Pyrimidine Synthesis Regulation

CPS-II activated by PRPP & ATP
Inhibited by UTP

Orotic Aciduria

Orotate
phosphoribosyltransferase &
OMP decarboxylase (UMP
Synthase)

Low activity = ↑ orotic acid

